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Identification of Torsional Modes in Complex Molecules Using Redundant Internal Coordinates: The Multistructural Method with Torsional Anharmonicity with a Coupled Torsional Potential and Delocalized Torsions

Wenqi Chen, Pengchao Zhang, Donald G. Truhlar, Jingjing Zheng,* and Xuefei Xu*

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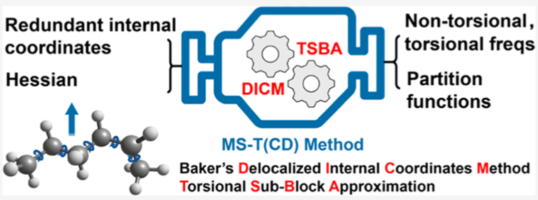

ABSTRACT: Identification of internal-rotation modes in the normal-mode analysis of complex molecules is important for accurately describing the thermodynamic properties and kinetics of complex molecules when it is necessary to treat the anharmonicity of torsions and the multiconformer anharmonicity caused by the internal rotations. However, identifying and distinguishing torsional modes are very challenging because they are coupled to one another. In this work, we present a new strategy to automatically identify torsional vibrations and separate them from the other vibrational modes. By combining a redundant-internal-coordinate auto-generation procedure with torsional projection techniques, we automate the procedure of identifying and separating the coupled torsions, and we show that we can obtain robust and consistent results with various reasonable definitions of redundant-internal-coordinate sets. This model has been implemented in a new development version of the *MSTor* program to reduce the user input needed for multistructural and torsional anharmonicity (MS-T) calculations. The new method is called multistructural and torsional anharmonicity with a coupled torsional potential and delocalized torsions ([MS-T(CD)]. As example applications, we consider MS-T(CD) calculations on three molecules (2-hexyl radical, *n*-propylbenzene, and 5-hydroperoxy-6-oxohexanoylperoxy radical) that have multiple rotors and that provide challenges to choosing good sets of nonredundant-internal coordinates, and we compare the performance of the new strategy to five other torsion identification methods. The new strategy is demonstrated to be efficient in separating the torsional and nontorsional elements in the Hessian matrix, as well as in providing reasonable projected nontorsional frequencies to be used for calculations of partition function and thermochemistry.

1. INTRODUCTION

Due to their significant anharmonicity, large-amplitude motions [such as low-frequency out-of-plane bends and internal rotations (torsions)] of complex molecules usually need special treatments for obtaining accurate partition functions (PFs), thermodynamic functions, and/or transition-state-theory rate constants. Anharmonicity of out-of-plane bending has been well addressed before.[1−5] It is more challenging to correct for the potential anharmonicity of torsions and for the multistructural anharmonicity caused by a large number of low-energy conformers generated by the flexible internal rotations around single bonds. Furthermore, torsions are often strongly coupled to one another and to bends and stretches. Feynman path integral methods[6−8] and vibrational configuration interaction[9,10] provide a straightforward way to include all anharmonicities in quantum mechanical PFs, but they are only practical for small molecules due to the computational cost. Therefore, a more practical strategy is desired for medium-to-large molecules.

Many methods to properly consider torsional anharmonicity have been developed based on Pitzer and Gwinn's seminal work in the 1940s,[11−13] including separable one-dimensional hindered rotor (1D-HR),[14,15] *n*-dimensional hindered rotor (*n*D-HR),[16] extended hindered rotor (EHR),[17] extended two-dimensional torsion method (E2DT),[18] and the multistructural and torsional method (MS-T).[19,21] Among these methods, the MS-T method has the advantage of balancing the computational costs, ease of use, and accuracy, and it has been used for thermochemistry and kinetic studies of various systems.[22−54] Specifically, by using a nonredundant-internal-coordinate representation and the full Kilpatrick−Pitzer treatment, the MS-T method enables one to describe torsions including kinetic and potential energy coupling of torsions and Coriolis coupling to rotation without calculating the fully coupled many-dimensional torsional potential energy surface. The

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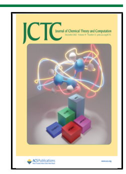

nonredundant-internal-coordinate representation plays an important role in the MS-T scheme because internal coordinates are used to separate the torsional modes from the overall vibrational Hessian. However, the use of nonredundant internal coordinates requires each torsion to be described by a single dihedral angle, and for this reason, the construction of a useful nonredundant representation is often a trial-and-error process. Improper choices of nonredundant-internal-coordinate sets may lead to erroneous torsional identifications, which subsequently affect the calculations of the coupled torsional frequencies and the coupled effective torsional barriers, thus finally giving inaccurate PFs and thermodynamic properties and/or rate constants. For example, Rissanen et al.[30] failed to find a reasonable effective torsional barrier for the 5-hydroperoxy-6-oxohexanoylperoxy radical (henceforth simply called the $C_6H_9O_6$ radical) with the MS-T method (details can be found in their Supporting Information). As discussed below, this is due to an improper choice of nonredundant internal coordinates that led to problematic torsional frequencies and barriers.

One drawback of using nonredundant internal coordinates to project coupled torsions is that the choice of a proper set of nonredundant coordinates is often not obvious and is difficult to automate for general cases. Our goal here is to identify torsions in normal-mode vibrations automatically and accurately, as well as to simplify the required input for MS-T calculations, and we here provide a new strategy to identify torsional modes using redundant internal coordinates. Recently, an iterative algorithm to obtain a set of well-chosen nonredundant internal coordinates for complex molecules was implemented in the *Q2DTor*[55] and *TorsiFlex*[56,57] programs. (The *TorsiFlex* program can create an input file for the *MSTor* program[58] to carry out MS-T calculations.) Our goal is that the method presented here should be simpler, more stable, and more robust as compared to the algorithm of *Q2DTor*. The method presented here separates torsions from other vibrations by projection in redundant internal coordinates and does not involve any iterative calculations. Furthermore, the method is made conformer-independent by starting from a set of highly redundant internal coordinates, and it does not include any assumptions about the natures of the modes except for the user identification of the single bonds whose internal rotations are considered.

The new algorithm presented here is used to improve the MS-T method that uses nonredundant internal coordinates. The method is based on Baker et al.'s delocalized-internal-coordinate method[59] and Zheng and Truhlar's torsion projection step in the original MS-T method.[21] This torsion projection step is called the torsional sub-block approximation in the present work. The redundant-internal-coordinate method was originally developed for geometry optimization, and it imposes a geometrical constraint on internal coordinates to separate constrained coordinates from the others during constrained optimization. Here, we use the geometrical constraint procedure to identify and separate the torsions from the other vibrations. The new strategy presented here circumvents the need to define nonredundant internal coordinates in the MS-T method, and it can straightforwardly separate any number of coupled torsions from the primitive Hessian.

Because the identification of torsions is a fundamental issue in torsional anharmonicity calculations,[60,61] several other methods have been proposed to separate torsions from overall normal-mode vibrations—either in an uncoupled approximation or in a coupled. In an uncoupled approximation, Pfaendtner et al. (2007) used the most similar harmonic normal mode to represent the corresponding torsions;[15] or as one option proposed by Vansteenkiste et al. (2006),[17] one can calculate the second derivative of the 1D internal-rotational potential at the equilibrium position as the torsional frequency. The coupled models usually project all torsions by projection matrices that are constructed by various methods. For example, Sharma et al. (2010) used torsional, translational, and rotational vectors to build a projection matrix in Cartesian coordinates;[62] Zheng and Truhlar (2013)[21] and Ayala and Schlegel (1998)[63] proposed multidimensional models for identifying the coupled torsional modes in nonredundant and redundant internal coordinates, respectively. We will compare our approach to these five previously developed ones. A glossary of acronyms for the various methods is given in Table 1.

Table 1. Glossary of Torsional Identification Schemes

type	acronym	authors
Uncoupled Approximations		
normal-mode approximation	NMA	Pfaendtner et al.[15]
second derivative of the torsional potential	1D-SDP	
Coupled Approximations		
Cartesian projection method	SRG	Sharma et al.[60]
redundant-internal-coordinate constraint method	AS	Ayala and Schlegel[61]
MS-T with a coupled torsional potential[a]	MS-T(C)	Zheng and Truhlar[21]
MS-T with a coupled torsional potential and delocalized torsions[a]	MS-T(CD)	this work

[a]MS-T denotes multistructural method with torsional anharmonicity.

The torsional identification process also has potential usefulness in potential energy distribution analysis[64] of vibrational spectra.

The present paper is organized as follows: Section 2 gives brief introductions to Baker et al.'s delocalized-internal-coordinate method, Zheng and Truhlar's torsional projection techniques, and the MS-T conformational-rotational-vibrational PFs; Section 3 describes the new method; and Sections 4 and 5 present the computational details and the results for the application of the new strategy to several cases. Section 6 is a summary.

2. METHODS

2.1. Redundant Internal Coordinates.

We begin with a brief introduction to Baker et al.'s delocalized-internal-coordinate method and its application in our strategy. More details can be found in refs 57 and 65.

For an arbitrary set of N primitive redundant internal coordinates $\boldsymbol{q}$, the Wilson $\mathbf{B}$ matrix[66] transforming Cartesian displacements $\delta\boldsymbol{x}$ to displacements $\delta\boldsymbol{q}$ in internal coordinates is denoted as $\mathbf{B}^{\text{prim}}$ with matrix elements $B_{ij}^{\text{Prim}} = \frac{\partial q_i}{\partial x_j}$. The displacements $\delta\boldsymbol{q}$ are obtained by

$$\delta\boldsymbol{q} = \mathbf{B}^{\text{prim}}\delta\boldsymbol{x} \tag{1}$$

If there are N_{atoms} atoms, then $\mathbf{B}^{\text{prim}}$ is a $N \times 3N_{\text{atoms}}$ matrix. Following Baker et al., we use the word "redundant" in two

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ways: a redundant set of coordinates is a set containing more than the F-independent internal coordinates (where F equals $3N_{\text{atoms}} - 6$ for nonlinear molecules or $3N_{\text{atoms}} - 5$ for linear molecules), and a redundant subset denotes the remaining coordinates after a nonredundant subset has been selected from the redundant set. Baker et al. separate a redundant set of N internal coordinates into a subset of F nonredundant coordinates and a subset of $(N - F)$ redundant coordinates by forming and diagonalizing the following $N \times N$ matrix

$$\mathbf{G} = \mathbf{B}^{\text{prim}}(\mathbf{B}^{\text{prim}})^{\text{T}} \tag{2}$$

The nonredundant subset is the F eigenvectors with nonzero eigenvalues, and the redundant subset has eigenvalues Λ_i equal to 0. The diagonalized matrix is called $\mathbf{G}^*$, and it satisfies the following eigenvalue equation

$$\mathbf{G}^*\mathbf{W} = \mathbf{W}\begin{pmatrix} \mathbf{\Lambda} & 0 \\ 0 & 0 \end{pmatrix} \tag{3}$$

where $\mathbf{\Lambda}$ is a diagonal matrix containing the nonzero Λ_i, and $\mathbf{W}$ is an $N \times N$ matrix whose first F columns are the $N \times F$ matrix $\mathbf{U}$ whose columns (called $\mathbf{U}_k$) are the nonredundant eigenvectors, and whose next $(N - F)$ columns are the $N \times (N - F)$ matrix $\mathbf{R}$ whose columns are the redundant eigenvectors.

In order to perform constrained optimization, Baker et al. separated specific internal coordinates from the full set of vibrations by Gram-Schmidt orthogonalization. We will follow the same procedure for obtaining the torsionally constrained eigenvector matrix $\mathbf{U}^{\text{t}}$. The procedure is as follows:

First, construct an initial torsion-only matrix $\mathbf{C}$, of dimension $N \times t$, which includes t vectors with the ith vector $\mathbf{C}_i$ corresponding to the ith target torsion. If n_i dihedrals in the primitive redundant internal coordinates can represent the ith torsion, the $\mathbf{C}_i$ vector is constructed with equal nonzero weights of $1/n_i$ for these dihedrals and zero weights for other internals. (Note that torsions are usually represented by dihedral angles, but dihedral angles are sometimes used for the coordinates of cyclic ring motion and out-of-plane vibrations. Here, only dihedrals related to the user-defined rotating single bonds are used in constructing the t vectors.)

Then, project the vectors of $\mathbf{C}$ onto the nonredundant eigenvector matrix $\mathbf{U}$ according to

$$\mathbf{C}_i^{\text{proj,unnorm}} = \sum_k [\mathbf{C}_i^{\text{T}}\mathbf{U}_k]\mathbf{U}_k \tag{4}$$

where $\mathbf{C}_i^{\text{proj,unnorm}}$ is $N \times 1$. Next, normalize $\mathbf{C}_i^{\text{proj,unnorm}}$ and call the result $\mathbf{C}_i^{\text{proj}}$. Then, form a matrix $\mathbf{V}$ of dimension $N \times (t + F)$ by

$$\mathbf{V} = [\mathbf{C}_i^{\text{proj}},\ \mathbf{U}_k](i = 1, ..., t;\ k = 1, ..., F) \tag{5}$$

Then, use Gram-Schmidt orthogonalization to orthogonalize $\mathbf{V}$, resulting in a new matrix $\mathbf{V}^{\text{o}}$ with the structure

$$\mathbf{V}^{\text{o}} = [\mathbf{C}^{\text{proj, o}},\ \mathbf{U}^{\text{o}},\ \mathbf{0}] \tag{6}$$

The matrix $\mathbf{V}^{\text{o}}$ is a combination of the $N \times t$ matrix $\mathbf{C}^{\text{proj,o}}$ consisting of t orthogonalized torsionally constrained vectors $\mathbf{C}_i^{\text{proj,o}}$, the $N \times (F - t)$ matrix $\mathbf{U}^{\text{o}}$ consisting of $(F - t)$ vectors that are orthogonal to each other and to the vectors $\mathbf{C}_i^{\text{proj,o}}$, and the $N \times t$ matrix $\mathbf{0}$ with t zero vectors. This procedure has removed t selected torsions from $\mathbf{U}^{\text{o}}$.

Next, the $N \times F$ dimension torsionally constrained eigenvector matrix $\mathbf{U}^{\text{t}}$ is constructed by combining the vectors of $\mathbf{U}^{\text{o}}$ with the original torsionally constrained vectors in $\mathbf{C}$

$$\mathbf{U}^{\text{t}} = [\mathbf{U}^{\text{o}},\ \mathbf{C}] \tag{7}$$

where $\mathbf{U}^{\text{t}}$ has dimension $N \times F$. Finally, we obtain the Wilson $\mathbf{B}$ matrix of dimension $F \times 3N_{\text{atoms}}$ by

$$\mathbf{B} = (\mathbf{U}^{\text{t}})^{\text{T}}\mathbf{B}^{\text{prim}} \tag{8}$$

With the $\mathbf{B}$ matrix constructed in this way, the vibrational normal-mode analysis is done by the standard Wilson GF method. It reproduces the normal-mode frequencies by diagonalizing the (translation−rotation)-free Hessian in mass-weighted Cartesian coordinates, as done in ref 67.

2.2. Torsional Frequencies and Barriers.

After the Wilson $\mathbf{B}$ matrix over the set of nonredundant torsionally constrained active internal coordinates is obtained by eq 8, the frequencies of torsional modes are obtained by using the MS-T(C) method[21] in which only the sub-blocks of torsional internal coordinates of the full Wilson $\mathbf{F}$ and $\mathbf{G}^{-1}$ matrices are used to calculate the frequencies. This includes the coupling of torsional modes to one another, but the coupling between torsional and nontorsional modes is ignored.[21] The details are given below.

The Wilson−Decius−Cross $\mathbf{G}$ matrix[64] and force constant matrix $\mathbf{F}$ for nonredundant torsionally constrained active internal coordinates are calculated, respectively, by

$$\mathbf{G} = \mathbf{B}\mathbf{u}\mathbf{B}^{\text{T}} \tag{9}$$

$$\mathbf{F} = \mathbf{A}^{\text{T}}\mathbf{F}^{\text{Cart}}\mathbf{A} = (\mathbf{u}\mathbf{B}^{\text{T}}\mathbf{G}^{-1})^{\text{T}}\mathbf{F}^{\text{Cart}}(\mathbf{u}\mathbf{B}^{\text{T}}\mathbf{G}^{-1}) \tag{10}$$

where $\mathbf{B}$ is an $F \times 3N_{\text{atoms}}$ matrix, $\mathbf{u}$ is taken as a diagonal matrix with the reciprocals of the atomic masses as the diagonal elements, $\mathbf{A}$ denotes the generalized inverse of the $\mathbf{B}$ matrix, and $\mathbf{F}^{\text{Cart}}$ is the original Hessian matrix in atomic Cartesian coordinates. Then, the full $\mathbf{F}$ and $\mathbf{G}^{-1}$ matrices can be represented in terms of submatrices as given below

$$\mathbf{F} = \begin{Vmatrix} \mathbf{F}^{11} & \mathbf{F}^{12} \\ \mathbf{F}^{21} & \mathbf{F}^{22} \end{Vmatrix}, \quad \mathbf{G}^{-1} = \begin{Vmatrix} \mathbf{Y}^{11} & \mathbf{Y}^{12} \\ \mathbf{Y}^{21} & \mathbf{Y}^{22} \end{Vmatrix} \tag{11}$$

where the submatrices with label 22 are for torsional internal coordinates. That is, $\mathbf{F}^{\text{tor}} = \mathbf{F}^{22}$ and $\mathbf{G}^{\text{tor}} = (\mathbf{Y}^{22})^{-1}$, both of which are $t \times t$ matrices, where t is the number of torsions. Then, projected torsional frequencies can be obtained through diagonalization of the $\mathbf{G}^{\text{tor}}\mathbf{F}^{\text{tor}}$ matrix

$$\mathbf{G}^{\text{tor}}\mathbf{F}^{\text{tor}}\mathbf{L} = \mathbf{L}\mathbf{\Lambda} \tag{12}$$

where $\mathbf{L}$ is the matrix of the generalized normal-mode eigenvectors, and $\mathbf{\Lambda}$ is the diagonal eigenvalue matrix. The square roots of the diagonal elements of $\mathbf{\Lambda}$ are the torsional vibrational frequencies.

In addition to the torsional frequencies, torsional barrier heights are also important parameters in the MS-T calculations, where they are used to formulate the reference torsional potential function. Zheng and Truhlar[21] proposed to calculate the coupled torsional barrier by scaling the $\mathbf{F}^{\text{tor}}$ matrix with a diagonal matrix $\mathbf{M}$

$$\tilde{\mathbf{F}}^{\text{tor}} = \mathbf{M}\mathbf{F}^{\text{tor}}\mathbf{M} \tag{13}$$

The matrix elements of $\mathbf{M}$ are given by $1/M_\tau$, where M_τ is the local periodicity of the specific torsion τ and they can be obtained by the Voronoi algorithm for the coupled torsions.

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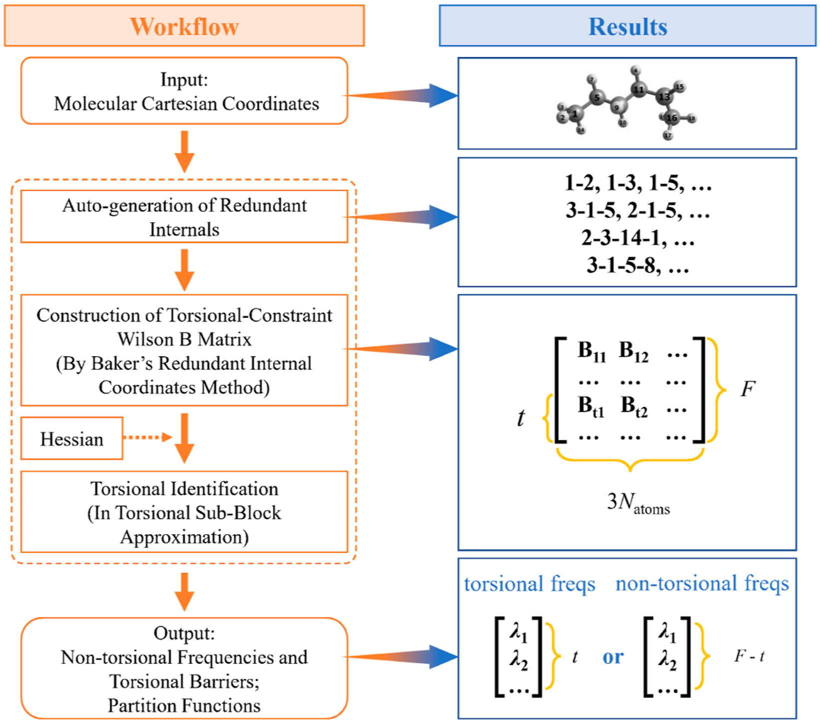

Figure 1. Workflow of the automatized redundant torsional identification.

The coupled barrier of torsion τ is $W_\tau = 2\tilde{\lambda}_\tau$, where $\tilde{\lambda}_\tau$ are the eigenvalues of the $\tilde{\mathbf{F}}^{\text{tor}}$ matrix. The local periodicity parameters M_τ are determined by the torsional angle distribution in the conformational space. Therefore, given a distribution of torsional conformers, the coupled torsional barrier heights only depend on the $\mathbf{F}^{\text{tor}}$ matrix, which is directly related to the choice of internal coordinates.

2.3. Nontorsional Frequencies. It is often the case that the magnitude of the matrix element associated with the coupling of a torsional coordinate to a nontorsional coordinate in the **F** and **G** matrices is larger than the magnitude of diagonal matrix elements of the torsional block.[21] Thus, simply removing the nontorsional part in the full Wilson **F** and $\mathbf{G}^{-1}$ matrices and ignoring the coupling between the torsional and nontorsional modes can lead to large errors in torsional frequencies. In contrast, the higher-frequency nontorsional frequencies are less affected by ignoring the torsion–nontorsion coupling elements because the diagonal matrix elements corresponding to nontorsional coordinates are often much larger than them. Hence, the MS-T method uses the nontorsional counterparts for torsional anharmonicity corrections instead of directly using the torsional frequencies. Detailed descriptions of "torsional anharmonicity corrections" in the MS-T method can be found in Section 2.4.

Similarly, the nontorsional frequencies are computed with the Wilson GF method using the torsionally constrained Wilson **B** matrix by removing the rows associated with torsional internal coordinates, $\mathbf{B}^{\text{non-tor}}$. The corresponding $\mathbf{G}^{\text{non-tor}}$ matrix and force constant matrix $\mathbf{F}^{\text{non-tor}}$ for nontorsional internal coordinates are calculated, respectively, by

$$\mathbf{G}^{\text{non-tor}} = \mathbf{B}^{\text{non-tor}}\mathbf{u}(\mathbf{B}^{\text{non-tor}})^{\text{T}} \tag{14}$$

$$\mathbf{F}^{\text{non-tor}} = (\mathbf{A}^{\text{non-tor}})^{\text{T}}\mathbf{F}^{\text{Cart}}\mathbf{A}^{\text{non-tor}} \tag{15}$$

where $\mathbf{B}^{\text{non-tor}}$ is a $(F - t) \times 3N_{\text{atoms}}$ matrix, and $\mathbf{A}^{\text{non-tor}}$ denotes the generalized inverse of the $\mathbf{B}^{\text{non-tor}}$ matrix. The nontorsional frequencies are computed as the square roots of the eigenvalues of the $\mathbf{G}^{\text{non-tor}}\mathbf{F}^{\text{non-tor}}$ matrix.

Note that the $\mathbf{F}^{\text{non-tor}}$ matrix here is not the same as the $\mathbf{F}^{11}$ matrix in eq 11. Although vectors in the torsionally constrained eigenvector matrix $\mathbf{U}^{\text{t}}$ are orthogonal to each other, vectors representing different internal coordinates in the torsionally constrained Wilson **B** matrix are not totally orthogonal to each other. Thus, the **B** and $\mathbf{B}^{\text{non-tor}}$ matrices can transform the $\mathbf{F}^{\text{Cart}}$ matrix differently.

2.4. MS-T Conformational–Rotational–Vibrational PFs. Complex molecules with multiple torsional degrees of freedom usually have multiple minima (conformers) on their potential energy surfaces, and thus—in addition to including the torsional anharmonicity—we also consider multistructural anharmonicity, which accounts for multiple conformers. The MS-T method[21] includes both multistructural and torsional anharmonicity corrections in PFs using either an uncoupled [MS-T(U)] or a coupled [MS-T(C)] torsional potential, where we prefer the coupled version in general. Here, because the main subject of this article is the coupling of torsions to one another, we only give a brief introduction to the MS-T calculations of conformational–rotational–vibrational PFs based on a coupled torsional potential ($Q^{\text{MS-T(C)}}$). More details about the MS-T method for PF calculations are given in refs 14 and 68.

The coupled conformational-torsional potential PF is

$$Q^{\text{MS-T(C)}} = \sum_{j=1}^{N} Q_j^{\text{SS-HO}} e^{-\beta U_j} \prod_{\eta=1}^{t} f_{j,\eta} \tag{16}$$

where N is the number of conformers, $Q_j^{\text{SS-HO}}$ is the single-structure (SS) rovibrational PF of conformer j in the rigid-rotor harmonic-oscillator (HO) approximation, β is $1/k_{\text{B}}T$, U_j is the potential energy of structure j relative to the lowest-energy conformer, and the $f_{j,\eta}$ factors correct the HO approximation for the torsional anharmonicity originating

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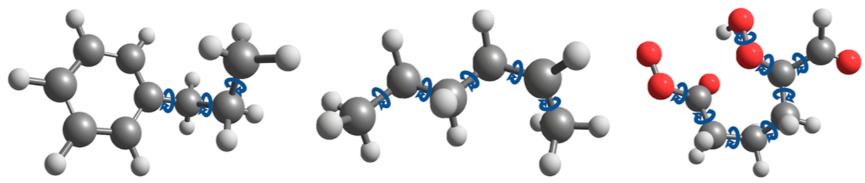

n-propylbenzene 2-hexyl $C_6H_9O_6$

Figure 2. Structures of *n*-propylbenzene, 2-hexyl radical, and $C_6H_9O_6$ radical.

from a set of coupled torsional motions indexed by η in the PF calculation for structure j.

Torsional anharmonicity correction factors $f_{j,\eta}$ are calculated through the reference Pitzer-Gwinn method.[11,14] Because all torsional motions are coupled, we approximate the product of the torsional anharmonicity factors at conformer j as follows

$$\prod_{\eta=1}^{t} f_{j,\eta} = (2\pi\hbar\beta)^{t/2} \frac{\prod_{m=1}^{F} \omega_{j,m}}{\prod_{\bar{\bar{m}}=1}^{F-t} \bar{\bar{\omega}}_{j,\bar{\bar{m}}}} \frac{\sqrt{\det \mathbf{D}_j}}{\prod_{\tau=1}^{t} M_{j,\tau}} \prod_{\eta=1}^{t} \exp\left(\frac{-\beta W_{j,\eta}^{(\mathrm{C})}}{2}\right) I_0\left(\frac{\beta W_{j,\eta}^{(\mathrm{C})}}{2}\right) \tag{17}$$

where $\omega_{j,m}$ and $\bar{\bar{\omega}}_{j,m}$ denote the normal-mode frequencies and nontorsional frequencies of conformer j, respectively, $\mathbf{D}_j$ represents the Kilpatrick−Pitzer torsional moment of inertia matrix[13,20] for conformer j, and $M_{j,\tau}$ and $W_{j,\eta}^{(\mathrm{C})}$ are the local periodicity and the effective torsional barrier associated with the coupled torsional motion η at structure j, respectively.

By using eqs 16 and 17, the MS-T method combines the normal-mode frequencies and nontorsional frequencies to obtain the overall torsional anharmonicity correction, and it avoids the direct use of torsional frequencies in PF calculations. This is preferred because they can be adversely affected by the neglect of the torsion−nontorsion couplings.

3. WORKFLOW OF TORSION-MODE IDENTIFICATION AND SEPARATION

The torsion-mode identification and separation method presented in this article (denoted as MS-T(CD) in Table 1) follows a three-step process: (1) generate primitive redundant internal coordinates using an automatic procedure proposed by Peng et al.,[69] which has been implemented in many computational chemistry programs; here we use the implementation in the Pilgrim program[70] as our reference; (2) transform the primitive redundant internal coordinates to a set of nonredundant torsionally constrained active internal coordinates by using Baker's method and construct the corresponding Wilson **B** matrix; and (3) calculate torsional frequencies, barriers, and nontorsional frequencies as discussed in Sections 2.2 and 2.3 using the torsional sub-block approximation.[21] The general workflow is shown in Figure 1.

4. COMPUTATIONAL DETAILS

We applied the proposed method to the three challenging cases: 2-hexyl radical, $C_6H_9O_6$ radical, and *n*-propylbenzene (see Figure 2 and the Supporting Information for the structures). To demonstrate the performance of the new MS-T(CD) method, the results are compared to those obtained by the other torsional identification methods mentioned in Section 1. Because there are no standard criteria for evaluating torsional identification approaches, we mainly used two criteria to judge the performance of these methods. One is that the calculated nontorsional frequencies should be close to their counterparts obtained by the standard normal-mode procedure, especially for the high-frequency normal modes; the other is that the calculated projected torsional frequencies ought to be relatively small and reasonable.

Because some methods yield only nontorsional frequencies or only torsional frequencies, the first criterion is applied for comparison between the SRG, MS-T(C), and MS-T(CD), while the second one is used for comparison between the NMA, 1D-SDP, MS-T(C), AS, and MS-T(CD).

We also compared the SS nontorsional PFs to illustrate the influence of different torsion-mode approximations on the PFs. Exhaustive conformer searching and the torsional-anharmonicity-corrected multistructural rovibrational PFs are also presented to show the robustness of our newly proposed method.

All electronic structure calculations were carried out by Gaussian 16.[71] Geometry optimization and normal-mode frequency analysis were calculated by using the M06-2X[72]/def2-TZVP[73] density functional method for *n*-propylbenzene and 2-hexyl radicals. To compare with ref 30 (details can be found in their Supporting Information), geometry optimization and normal-mode frequency analysis for the $C_6H_9O_6$ radical were performed by the ωB97X-D[74]/aug-cc-pVTZ[75] method used in ref 30. We did not scale the calculated frequencies in these tests.

The torsional identifications were performed by using the *Mesmer 6.0* program[76] for the SRG model, the *MSTor 2017* program[58] for the MS-T(C) model, *Gaussian 16* software[69] with the keyword *"freq = hinderedrotor"* for the AS method, and *Gaussian 16* software with relaxed 1D scans for the 1D-SDP method. The reduced moments of inertia for torsions required by the 1D-SDP method were calculated using Kilpatrick and Pitzer's general *N*-dimensional treatments,[13] which have been implemented in the *MSTor 2017* program.

5. RESULTS AND DISCUSSION

5.1. Geometries. We chose the 2-hexyl radical, *n*-propylbenzene, and $C_6H_9O_6$ radical as our test cases because they all have multiple torsional modes and these modes are strongly coupled with each other and even coupled with nontorsional vibrational modes. Another important reason to choose the $C_6H_9O_6$ radical is to compare the results obtained by our newly proposed method with results of Rissanen et al. in ref 30, where they pointed out that torsional sub-block approximation in nonredundant internal coordinates [denoted as MS-T(C) in Table 1] failed to produce reasonable torsional barriers.

There are 3, 5, and 8 rotatable bonds in *n*-propylbenzene, 2-hexyl radical, and $C_6H_9O_6$ radical, respectively. Molecular structures are shown in Figure 2, and the rotatable bonds are labeled with curved arrows. These torsions generate multiple

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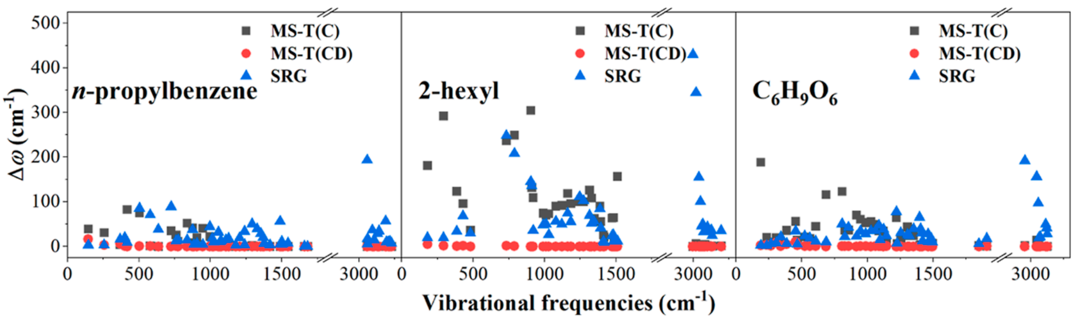

Figure 3. Deviations ($\Delta\omega$) of nontorsional vibrational frequencies obtained using different methods from the standard normal-mode frequencies for *n*-propylbenzene, 2-hexyl radical, and $C_6H_9O_6$ radical. The MS-T(C) calculations in this figure used an inappropriate nonredundant-internal-coordinate set.

Table 2. Projected Torsional Frequencies[a]

species	method	projected torsional frequencies (cm^{-1})				
2-hexyl radical	NMA	50	90	113	123	248
	1D-SDP	58	102	115	119	221
	MS-T(C)[b]	79	133	213	836	1350
	MS-T(C)[c]	56	95	122	136	252
	AS	56	95	122	136	252
	MS-T(CD)	56	95	122	136	252
n-propylbenzene	NMA	45	86	256		
	1D-SDP	44	91	220		
	MS-T(C)[b]	46	108	254		
	MS-T(C)[c]	46	108	254		
	AS	46	108	254		
	MS-T(CD)	46	108	254		
$C_6H_9O_6$ radical	NMA	42	49	80	96	133
		146	186	427		
	1D-SDP	91	73	101	72	63
		67	137	406		
	MS-T(C)[b]	93	128	180	214	268
		308	793	1407		
	MS-T(C)[c]	45	53	86	103	140
		150	212	435		
	AS	45	53	86	103	140
		150	212	435		
	MS-T(CD)	45	53	86	103	140
		150	212	435		

[a]For this table, the nonredundant internal coordinates for the MS-T(C) calculations are set manually; the internal coordinate sets used for these calculations are listed in the Supporting Information. [b]These MS-T(C) calculations used an inappropriate nonredundant-internal-coordinate set. [c]These MS-T(C) calculations used a well-chosen nonredundant-internal-coordinate set.

conformers that can introduce considerable multistructural anharmonicity in calculations of thermochemical properties and dynamics. However, in the current study, we first focused on the torsional identification and determination of torsional and/or nontorsional frequencies and torsional barriers, and any reasonable stable conformer can be used for this first part of the study (the selected geometries are given in the Supporting Information). The overall PFs and the exhaustive conformer searching results are given in Section 5.5.

5.2. Nontorsional Frequencies. The nontorsional vibrational frequencies were calculated using the new method in redundant internal coordinates [MS-T(CD)] and using methods presented by Sharma et al. (SRG) and Zheng and Truhlar [MS-T(C)], respectively. The deviations of these frequencies from the standard normal-mode frequencies are shown in Figure 3 for comparison.

Figure 3 shows that the nontorsional frequencies computed using the SRG and MS-T(C) methods show significant deviations from the standard normal-mode frequencies, even for some very-high-frequency modes. We note that Gao et al.[77] also reported large discrepancies between the SRG results and their normal-mode counterparts at high frequencies for *n*-butane. The large deviation of the MS-T(C) nontorsional frequencies from standard normal-mode frequencies can be caused by a bad choice of the nonredundant internal coordinates. The MS-T(C) method uses nonredundant internal coordinates with each torsional motion represented by a single dihedral angle. However, it is very hard to construct an appropriate set of nonredundant internal coordinates for a complex molecule if each torsion is described by only one dihedral angle. Redundant internal coordinates are usually better choices for describing the molecule vibrations, as has been widely recognized by their use for geometry optimiza-

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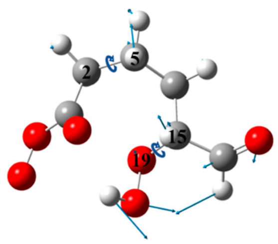
tions. The MS-T(CD) method in this present study can avoid the problems caused in the original MS-T calculations by using inappropriate nonredundant internal coordinates. As shown in Figure 3, the MS-T(CD) method reproduces the standard frequencies of the nontorsional modes quite well.

Note that our method is not sensitive to the choice of primitive redundant internal coordinates, that is, one can obtain consistent and accurate results with different primitive redundant-internal sets. Comparisons of calculations with different redundant-internal-coordinate sets can be found in the Supporting Information. As demonstrated by these examples, redundant internal coordinates help to eliminate the errors in calculating the vibrational frequencies when separating the torsional modes and nontorsional modes.

5.3. Projected Torsional Frequencies. The projected torsional frequencies calculated with the methods mentioned above (excluding the SRG model because it can only generate nontorsional frequencies) are given in Table 2. The table shows that although the NMA and 1D-SDP uncoupled approximations sometimes give reasonable results, they may encounter problems in treating large flexible systems. For example, various torsional motions may be heavily mixed in a normal mode as shown in Figure 4; therefore, it is usually quite difficult to employ the NMA model by assigning a normal mode as a specific torsion. Furthermore, the 1D-SDP method is unable to capture the couplings of torsional modes because it simply uses the uncoupled torsional potential energy surface obtained through separate relaxed 1D scans along torsion angles. For complex systems with strong torsional couplings, for example, the $C_6H_9O_6$ radical shown in Figure 4, it is hard to get reasonable torsional frequencies by using the 1D models to separate the torsional motions. Thus, in these cases, we recommend solving this identification problem in a multi-dimensional and coupled way, such as by the MS-T(C) with appropriate nonredundant internal coordinates, by Ayala and Schlegel's method, or by the MS-T(CD) method proposed here.



Normal Mode #7
Freq: 186 cm^{-1}

Figure 4. Displacement vectors for a normal mode of the $C_6H_9O_6$ radical. Torsions around bond 2−5 and bond 15−19 are heavily mixed in this normal mode.

The MS-T(C) results of complex molecules are sensitive to the selection of the nonredundant-internal-coordinate sets. Table 2 shows that some of the MS-T(C) results are very different from those obtained by the AS and our MS-T(CD) methods due to the inappropriate selection of the non-redundant internal coordinates. We held the opinion that if given a well-chosen and appropriate torsionally constrained nonredundant-internal-coordinate set, the MS-T(C) would produce results comparable to those obtained through redundant internals [i.e., the MS-T(CD) model proposed in this work]. To examine this, we recalculated projected torsional frequencies with the MS-T(C) method and a set of well-chosen nonredundant internals and showed them in Table 2 for comparison. It shows that with an appropriate choice of nonredundant internal coordinates, the MS-T(C) method can reproduce the results of the MS-T(CD) method. However, it is always a trial-and-error task to find a set of appropriate nonredundant internal coordinates and it is also not clear if such a set of coordinates always exist for any given system. Thus, it is a motivation to develop the new method to eliminate this difficulty and uncertainty.

Table 2 shows that our torsional frequencies calculated using redundant internal coordinates [MS-T(CD)] are in perfect agreement with those calculated via the AS model. Even though the calculation procedures and the redundant-internal-coordinate sets are completely different, the same results indicate that both models successfully minimize the couplings between torsional and nontorsional modes and identify torsions correctly.

Rissanen et al.[30] mentioned that they obtained problematic torsional barriers for the $C_6H_9O_6$ radical, and we also obtained such problematic results for this radical when using the MS-T(C) method with an inappropriate internal-coordinate set. However, as shown in Table 3, by using the MS-T(CD) method proposed in this work, the coupled torsional barriers

Table 3. Effective Torsional Barriers Calculated with Two Coupled Torsional Identification Models

species	method	torsional barriers (kcal/mol)[a]				
2-hexyl radical	MS-T(C)[b]	2.69	5.94	13.75	76.60	93.34
	MS-T(C)[c]	0.73	2.38	2.96	6.46	8.91
	MS-T(CD)	0.73	2.38	2.96	6.46	8.91
n-propylbenzene	MS-T(C)[b]	3.13	4.43	9.73		
	MS-T(C)[c]	3.10	3.97	9.70		
	MS-T(CD)	3.10	3.97	9.70		
$C_6H_9O_6$ radical	MS-T(C)[b]	17.1	32.1	59.8	124	206
		343	3162	13094		
	MS-T(C)[c]	7.04	14.1	19.8	27.0	48.3
		69.1	171	431		
	MS-T(CD)	7.04	14.1	19.8	27.0	48.3
		69.1	171	431		

[a]Torsional barriers are listed in ascending order. [b]These MS-T(C) calculations used an inappropriate nonredundant-internal-coordinate set. [c]These MS-T(C) calculations used a well-chosen nonredundant-internal-coordinate set.

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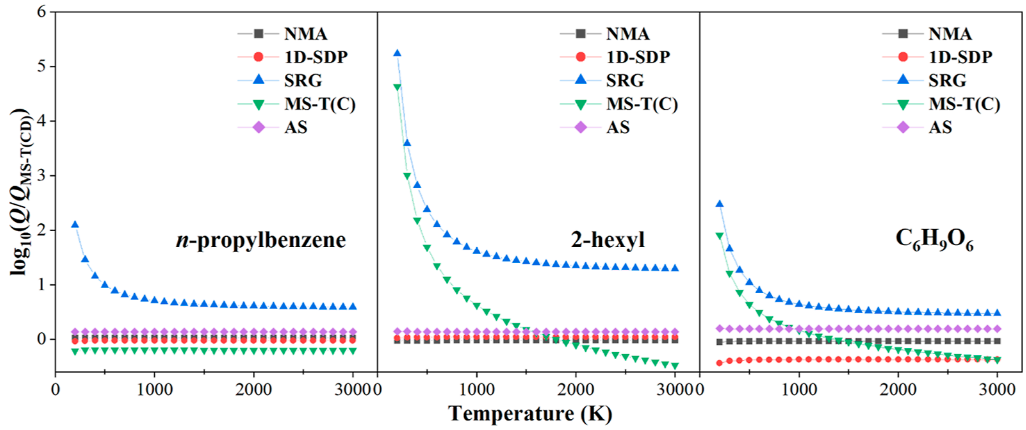

Figure 5. Logarithms of ratios of nontorsional PFs to those calculated by the MS-T(CD) method as functions of temperature. Inappropriate nonredundant internal coordinates are used in the calculations with the MS-T(C) model.

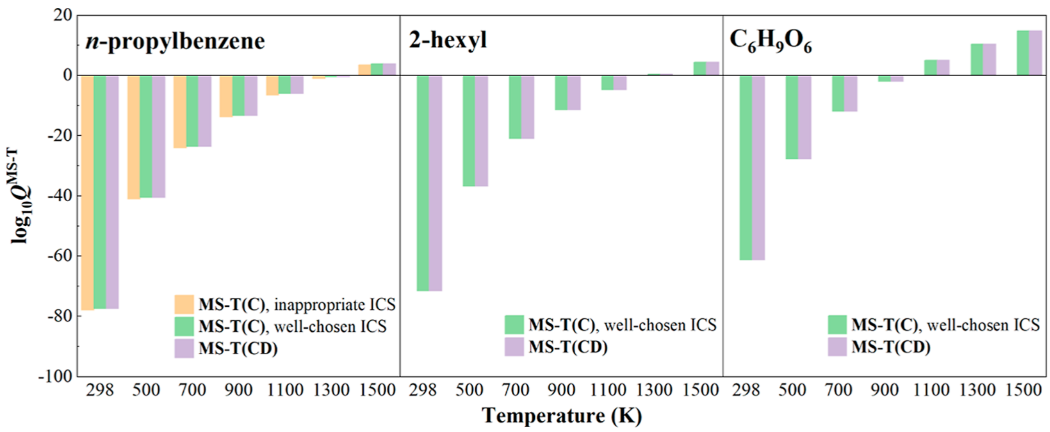

Figure 6. Multistructural rovibrational PF $Q^{MS\text{-}T}$ as a function of temperature. The nonredundant internal coordinates used in the MS-T(C) calculations are provided in the Supporting Information.

become much more reasonable and can be used for further calculations of thermochemical properties. The torsional barrier heights given in Table 3 are "effective" torsional barriers associated with the coupled potential; they are intermediate parameters in the algorithm rather than estimates of the actual barriers. A large effective torsional barrier need not lead to an incorrect PF. We previously said,[21] "We emphasize that these effective barriers do not correspond to specific features of the true potential energy surface but instead enable us to model the accessible phase space in the vicinity of structure *j*." The barriers only affect the transition behavior for the PFs from the low-temperature limit (quantum harmonic approximation) to high-temperature limit (classical torsion), and the MS-T(C) and MS-T(CD) methods both give the correct low-T and high-T limits. Using effective torsional barriers not only saves computational cost but also eliminates the manual effort of optimizing and assigning the torsional saddle points corresponding to specific structure-to-structure paths, and the latter is almost impossible for complex molecules with strongly coupled torsions and many conformational structures.

5.4. Nontorsional PFs at Various Temperatures. Here, we compared the nontorsional vibrational PFs calculated by various methods over a wide range of temperatures. For the NMA, SRG, MS-T(C), and MS-T(CD) methods, the PFs are computed directly using nontorsional vibrational frequencies, and for the 1D-SDP and AS methods, they are derived from the ratio of full vibrational HO PFs to the PFs calculated by projected torsional frequencies. The results are shown in Figure 5.

Figure 5 shows that the nontorsional PFs calculated with the AS model and the strategy proposed in this work do not equal each other. This is because the couplings between torsions and nontorsional modes cannot be completely eliminated, even in the redundant-internal-coordinate set.

It is not surprising that the blue (SRG) and green [MS-T(C)] curves in Figure 5 deviate significantly from zero because they produce questionable nontorsional frequencies. Note that the reason that the MS-T(C) method failed here is because we used an inappropriate set of nonredundant internal coordinates for these cases. In the original article describing Zheng and Truhlar's torsional projection method, they suggested experimenting with different sets of internals to find the best one.

The black (NMA) and red (1D-SDP) lines in Figure 5 show that treating the torsional anharmonicity in an uncoupled way may sometimes yield satisfactory nontorsional PFs, although the 1D-SDP method gives poor results for the $C_6H_9O_6$ radical. It is not clear if the good agreement sometimes obtained is due to a fortuitous cancellation of errors.

5.5. Comparison of MS-T PFs Calculated with the MS-T(C) and MS-T(CD) Methods. The MS-T(CD) method proposed here has been implemented in the development version of the *MSTor* program, and it provides an opportunity

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to examine the effect of the torsional identification on the calculated overall PFs with the MS-T method. Therefore, we compared the MS-T conformational−rovibrational PFs calculated with both the MS-T(CD) torsional identification method using redundant internal coordinates and the previous MS-T(C) method with a set of specified nonredundant internal coordinates. The details of MS-T calculations are given in the Supporting Information.

Results are given in Figure 6, which does not give the PFs for the 2-hexyl radical and $C_6H_9O_6$ radical calculated by the MS-T(C) method with the inappropriate nonredundant internals as the poor-chosen nonredundant-internal-coordinate set leads to numerical problems in calculations of PFs in these cases. This circumstance is also stressed in Rissanen's paper.[30] As seen in Figure 6, the PFs calculated by the MS-T(CD) method agree well with those obtained by the MS-T(C) method when the MS-T(C) results are based on well-chosen nonredundant internal coordinates. We regard the results obtained by the new MS-T(CD) method with redundant-internal-coordinate sets as the optimal solutions. Thus, with the redundant-internal-coordinate scheme proposed in this work, the MS-T method becomes more convenient and robust.

6. CONCLUDING REMARKS

We presented a new torsional identification approach based on Baker et al.'s delocalized redundant internal coordinates and on Zheng and Truhlar's torsional projection method. One difficulty in the original MS-T method with nonredundant coordinates is that having a unique set of nonredundant coordinates may work well for some conformers but not for others, a problem that is more likely to occur in systems with a large number of conformations. This problem is avoided by using the MS-T(CD) method that we propose here. In the MS-T(CD) method, instead of directly constructing the Wilson B matrix from the user-defined nonredundant coordinates, which may be influenced by conformational differences, the torsionally constrained Wilson B matrix is automatically constructed using Baker's delocalized strategy with a set of highly redundant primitive internal coordinates that works well for all conformers.

The performance of the new method has been examined by application to three challenging cases (2-hexyl radical, *n*-propylbenzene, and $C_6H_9O_6$ radical), each of which has multiple coupled internal rotations, and we compared the performance with five other torsional treatment methods in terms of their torsional and nontorsional frequencies, SS nontorsional PF, and/or conformational−rovibrational PFs, including both multistructural and torsional anharmonicity corrections. The results show the promising performance of the new method. This approach has been implemented in a development version of the *MSTor* program. Combined with the automatic generation procedure for internal coordinates, the use of the new torsional identification method greatly simplifies the user input of MS-T calculations by automated identification and separation of the coupled torsions, and it helps users to obtain robust and consistent results.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.2c00952.

Cartesian coordinates and ChemDraw structures; inappropriate nonredundant internal coordinates of *n*-propylbenzene, 2-hexyl, and $C_6H_9O_6$ for the MS-T(C) calculations; well-chosen nonredundant internal coordinates of *n*-propylbenzene, 2-hexyl radical, and $C_6H_9O_6$ radical for the MS-T(C) calculations; comparison of the MS-T(CD) calculations with different redundant-internal-coordinate sets; details of the exhaustive conformer searching procedure for the MS-T(C) and MS-T(CD) calculations; and Cartesian coordinates of *n*-propylbenzene and 2-hexyl conformers (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jingjing Zheng − *Gaussian, Inc., Wallingford, Connecticut 06492, United States*; Email: zheng@gaussian.com

Xuefei Xu − *Center for Combustion Energy, Department of Energy and Power Engineering, and Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Tsinghua University, Beijing 100084, China*; orcid.org/0000-0002-2009-0483; Email: xuxuefei@tsinghua.edu.cn

Authors

Wenqi Chen − *Center for Combustion Energy, Department of Energy and Power Engineering, and Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Tsinghua University, Beijing 100084, China*; orcid.org/0000-0002-1953-8383

Pengchao Zhang − *Center for Combustion Energy, Department of Energy and Power Engineering, and Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Tsinghua University, Beijing 100084, China*; orcid.org/0000-0002-6556-6588

Donald G. Truhlar − *Department of Chemistry, Chemical Theory Center, and Minnesota Supercomputing Institute, University of Minnesota, Minneapolis, Minnesota MN55455-0431, United States*; orcid.org/0000-0002-7742-7294

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jctc.2c00952

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Garrett, B. C.; Truhlar, D. G. Importance of quartic anharmonicity for bending partition functions in transition-state theory. *J. Phys. Chem.* **1979**, *83*, 1915−1924.

(2) Natanson, G. A. Separation and semiclassical quantization of bending motion near linear geometries of a triatom. *J. Chem. Phys.* **1990**, *93*, 6589−6606.

(3) Garrett, B. C.; Truhlar, D. G. Critical tests of variational transition state theory and semiclassical tunneling methods for hydrogen and deuterium atom transfer reactions and use of the semiclassical calculations to interpret the overbarrier and tunneling dynamics. *J. Phys. Chem.* **1991**, *95*, 10374−10379.

(4) Wu, J.; Gao, L. G.; Ren, W.; Truhlar, D. G. Anharmonic kinetics of the cyclopentane reaction with hydroxyl radical. *Chem. Sci.* **2020**, *11*, 2511−2523.

(5) Chen, W.; Guo, X.; Chen, L.; Zhang, R.; Li, Y.; Feng, H.; Xu, X.; Zhang, X. A kinetics study on hydrogen abstraction reactions of cyclopentane by hydrogen, methyl, and ethyl radicals. *Phys. Chem. Chem. Phys.* **2021**, *23*, 7333−7342.

(6) Feynman, R. P.*Statistical Mechanics: A Set of Lectures*; Basic Books, Benjamin: Reading, MA, 1972.

(7) Lynch, V. A.; Mielke, S. L.; Truhlar, D. G. High-precision quantum thermochemistry on nonquasiharmonic potentials: Converged path-integral free energies and a systematically convergent family of generalized Pitzer− Gwinn approximations. *J. Phys. Chem. A* **2005**, *109*, 10092−10099.

(8) Feynman, R. P.; Hibbs, A. R.*Quantum Mechanics and Path Integrals*; McGraw-Hill: New York, 1965.

(9) Carter, S.; Bowman, J. M.; Handy, N. C. Extensions and tests of "multimode": A code to obtain accurate vibration/rotation energies of many-mode molecules. *Theor. Chem. Acc.* **1998**, *100*, 191−198.

(10) Mielke, S. L.; Chakraborty, A.; Truhlar, D. G. Vibrational configuration interaction using a tiered multimode scheme and tests of approximate treatments of vibrational angular momentum coupling: A case study for methane. *J. Phys. Chem. A* **2013**, *117*, 7327−7343.

(11) Pitzer, K. S.; Gwinn, W. D. Energy levels and thermodynamic functions for molecules with internal rotation I. Rigid frame with attached tops. *J. Chem. Phys.* **1942**, *10*, 428−440.

(12) Pitzer, K. S. Energy levels and thermodynamic functions for molecules with internal rotation: II. Unsymmetrical tops attached to a rigid frame. *J. Chem. Phys.* **1946**, *14*, 239−243.

(13) Kilpatrick, J. E.; Pitzer, K. S. Energy levels and thermodynamic functions for molecules with internal rotation. III. Compound rotation. *J. Chem. Phys.* **1949**, *17*, 1064−1075.

(14) Ellingson, B. A.; Lynch, V. A.; Mielke, S. L.; Truhlar, D. G. Statistical thermodynamics of bond torsional modes: Tests of separable, almost-separable, and improved Pitzer−Gwinn approximations. *J. Chem. Phys.* **2006**, *125*, 084305.

(15) Pfaendtner, J.; Yu, X.; Broadbelt, L. J. The 1-D hindered rotor approximation. *Theor. Chem. Acc.* **2007**, *118*, 881−898.

(16) Speybroeck, V. V.; Vansteenkiste, P.; Neck, D. V.; Waroquier, M. Why does the uncoupled hindered rotor model work well for the thermodynamics of *n*-alkanes? *Chem. Phys. Lett.* **2005**, *402*, 479−484.

(17) Vansteenkiste, P.; Van Neck, D.; Van Speybroeck, V.; Waroquier, M. An extended hindered-rotor model with incorporation of Coriolis and vibrational-rotational coupling for calculating partition functions and derived quantities. *J. Chem. Phys.* **2006**, *124*, 044314.

(18) Simón-Carballido, L.; Bao, J. L.; Alves, T. V.; Meana-Pañeda, R.; Truhlar, D. G.; Fernández-Ramos, A. Anharmonicity of coupled torsions: The extended two-dimensional torsion method and its use to assess more approximate methods. *J. Chem. Theory Comput.* **2017**, *13*, 3478−3492.

(19) Zheng, J.; Yu, T.; Papajak, E.; Alecu, I. M.; Mielke, S. L.; Truhlar, D. G. Practical methods for including torsional anharmonicity in thermochemical calculations on complex molecules: The internal-coordinate multi-structural approximation. *Phys. Chem. Chem. Phys.* **2011**, *13*, 10885.

(20) Zheng, J.; Mielke, S. L.; Clarkson, K. L.; Truhlar, D. G. MSTor: A program for calculating partition functions, free energies, enthalpies, entropies, and heat capacities of complex molecules including torsional anharmonicity. *Comput. Phys. Commun.* **2012**, *183*, 1803−1812.

(21) Zheng, J.; Truhlar, D. G. Quantum thermochemistry: Multistructural method with torsional anharmonicity based on a coupled torsional potential. *J. Chem. Theory Comput.* **2013**, *9*, 1356−1367.

(22) Yu, T.; Zheng, J.; Truhlar, D. G. Statistical thermodynamics of the isomerization reaction between *n*-heptane and isoheptane. *Phys. Chem. Chem. Phys.* **2012**, *14*, 482−494.

(23) Xu, X.; Yu, T.; Papajak, E.; Truhlar, D. G. Multistructural variational transition state theory: Kinetics of the hydrogen abstraction from carbon-2 of 2-methyl-1-propanol by hydroperoxyl radical including all structures and torsional anharmonicity. *J. Phys. Chem. A* **2012**, *116*, 10480−10487.

(24) Seal, P.; Papajak, E.; Yu, T.; Truhlar, D. G. Statistical thermodynamics of 1-butanol, 2-methyl-1-propanol, and butanal. *J. Chem. Phys.* **2012**, *136*, 034306.

(25) Seal, P.; Papajak, E.; Truhlar, D. G. Kinetics of the hydrogen abstraction from carbon-3 of 1-butanol by hydroperoxyl radical: Multi-structural variational transition-state calculations of a reaction with 262 conformations of the transition state. *J. Phys. Chem. Lett.* **2012**, *3*, 264−271.

(26) Papajak, E.; Seal, P.; Xu, X.; Truhlar, D. G. Thermochemistry of radicals formed by hydrogen abstraction from 1-butanol, 2-methyl-1-propanol, and butanal. *J. Chem. Phys.* **2012**, *137*, 104314.

(27) Alecu, I. M.; Zheng, J.; Papajak, E.; Yu, T.; Truhlar, D. G. Biofuel combustion. Energetics and kinetics of hydrogen abstraction from carbon-1 in *n*-butanol by the hydroperoxyl radical calculated by coupled cluster and density functional theories and multistructural variational transition-state theory with multidimensional tunneling. *J. Phys. Chem. A* **2012**, *116*, 12206−12213.

(28) Xu, X.; Papajak, E.; Zheng, J.; Truhlar, D. G. Multi-structural variational transition state theory: Kinetics of the 1, 5-hydrogen shift isomerization of the 1-butoxyl radical including all structures and torsional anharmonicity. *Phys. Chem. Chem. Phys.* **2012**, *14*, 4204−4216.

(29) Yu, T.; Zheng, J.; Truhlar, D. G. Multipath variational transition state theory: Rate constant of the 1, 4-hydrogen shift isomerization of the 2-cyclohexylethyl radical. *J. Phys. Chem. A* **2012**, *116*, 297−308.

(30) Rissanen, M. P.; Kurtén, T.; Sipilä, M.; Thornton, J. A.; Kangasluoma, J.; Sarnela, N.; Junninen, H.; Jørgensen, S.; Schallhart, S.; Kajos, M. K.; Taipale, R.; Springer, M.; Mentel, T. F.; Ruuskanen, T.; Petäjä, T.; Worsnop, D. R.; Kjaergaard, H. G.; Ehn, M. The formation of highly oxidized multifunctional products in the ozonolysis of cyclohexene. *J. Am. Chem. Soc.* **2014**, *136*, 15596−15606.

(31) Seal, P.; Truhlar, D. G. Large entropic effects on the thermochemistry of silicon nanodusty plasma constituents. *J. Am. Chem. Soc.* **2014**, *136*, 2786−2799.

(32) Zheng, J.; Meana-Pañeda, R.; Truhlar, D. G. Prediction of experimentally unavailable product branching ratios for biofuel combustion: The role of anharmonicity in the reaction of isobutanol with OH. *J. Am. Chem. Soc.* **2014**, *136*, 5150−5160.

(33) Alecu, I. M.; Marshall, P. Computational Study of the Thermochemistry of $N_2O_5$and the Kinetics of the Reaction N_2O_5 + H_2O → 2 HNO_3. *J. Phys. Chem. A* **2014**, *118*, 11405−11416.

(34) Seal, P.; Zheng, J.; Truhlar, D. G. Entropic effects on the free energies of clusters in silane plasmas. *J. Phys. Chem. C* **2015**, *119*, 10085−10101.

(35) Bao, J. L.; Seal, P.; Truhlar, D. G. Nanodusty plasma chemistry: A mechanistic and variational transition state theory study of the initial steps of silyl anion−silane and silylene anion−silane polymerization reactions. *Phys. Chem. Chem. Phys.* **2015**, *17*, 15928−15935.

(36) Zheng, J.; Oyedepo, G. A.; Truhlar, D. G. Kinetics of the hydrogen abstraction reaction from 2-butanol by OH radical. *J. Phys. Chem. A* **2015**, *119*, 12182−12192.

(37) Bao, J. L.; Sripa, P.; Truhlar, D. G. Path-dependent variational effects and multidimensional tunneling in multi-path variational transition state theory: Rate constants calculated for the reactions of HO_2 with tert-butanol by including all 46 paths for abstraction at C and all six paths for abstraction at O. *Phys. Chem. Chem. Phys.* **2016**, *18*, 1032−1041.

(38) Li, X.; You, X.; Law, C. K.; Truhlar, D. G. Kinetics and branching fractions of the hydrogen abstraction reaction from methyl butenoates by H atoms. *Phys. Chem. Chem. Phys.* **2017**, *19*, 16563−16575.

(39) Xing, L.; Bao, J. L.; Wang, Z.; Zhang, F.; Truhlar, D. G. Degradation of Carbonyl Hydroperoxides in the Atmosphere and in Combustion. *J. Am. Chem. Soc.* **2017**, *139*, 15821−15835.

(40) Xing, L.; Bao, J. L.; Wang, Z.; Wang, X.; Truhlar, D. G. Relative rates of hydrogen shift isomerizations depend strongly on multiple-structure anharmonicity. *J. Am. Chem. Soc.* **2018**, *140*, 17556−17570.

(41) Gao, L. G.; Zheng, J.; Fernández-Ramos, A.; Truhlar, D. G.; Xu, X. Kinetics of the methanol reaction with OH at interstellar, atmospheric, and combustion temperatures. *J. Am. Chem. Soc.* **2018**, *140*, 2906−2918.

(42) Xing, L.; Bao, J. L.; Wang, Z.; Wang, X.; Truhlar, D. G. Hydrogen shift isomerizations in the kinetics of the second oxidation mechanism of alkane combustion. Reactions of the hydroperoxypentylperoxy OOQOOH radical. *Combust. Flame* **2018**, *197*, 88−101.

(43) Guan, Y.; Lou, J.; Liu, R.; Ma, H.; Song, J. Thermodynamic properties of the methylmethoxy radical with intricate treatment of two-dimensional hindered internal rotations. *J. Chem. Eng. Data* **2018**, *63*, 3640−3649.

(44) Monge-Palacios, M.; Rissanen, M. P.; Wang, Z.; Sarathy, S. M. Theoretical kinetic study of the formic acid catalyzed Criegee intermediate isomerization: Multistructural anharmonicity and atmospheric implications. *Phys. Chem. Chem. Phys.* **2018**, *20*, 10806−10814.

(45) Roberto-Neto, O.; Alves, T. V. Multipath VTST rate constants for D+ methyl formate reactions: Importance of torsional anharmonicity and conformational flexibility for combustion chemistry. *Chem. Phys. Lett.* **2018**, *711*, 132−137.

(46) Guan, Y.; Lou, J.; Ma, H.; Song, J. Investigation on the thermal dissociation of vinyl nitrite with a saddle point involved. *ACS Omega* **2019**, *4*, 16052−16061.

(47) Shang, Y.; Ning, H.; Shi, J.; Wang, H.; Luo, S.-N. Benchmarking dual-level MS-Tor and DLPNO-CCSD (T) methods for H-abstraction from methyl pentanoate by an OH radical. *Phys. Chem. Chem. Phys.* **2019**, *21*, 20857−20867.

(48) Wu, J.; Gao, L. G.; Ning, H.; Ren, W.; Truhlar, D. G. Direct dynamics of a large complex hydrocarbon reaction system: The reaction of OH with exo-tricyclodecane (the main component of Jet Propellant-10). *Combust. Flame* **2020**, *216*, 82−91.

(49) Duan, Y.; Monge-Palacios, M.; Grajales-Gonzalez, E.; Han, D.; Møller, K. H.; Kjaergaard, H. G.; Sarathy, S. M. Oxidation kinetics of n-pentanol: A theoretical study of the reactivity of the 1-hydroxy-1-peroxypentyl radical. *Combust. Flame* **2020**, *219*, 20−32.

(50) Duan, Y.; Huang, Z.; Han, D. Theoretical kinetics of hydrogen abstraction reactions from propanol isomers by hydroperoxyl radical: Implication for combustion modeling. *Combust. Flame* **2021**, *231*, 111495.

(51) Li, W.; Li, J.; Ning, H.; Shang, Y.; Luo, S.-N. Multistructural variational reaction kinetics of the simplest unsaturated methyl ester: H-abstraction from methyl acrylate by H, OH, CH_3, and HO_2 radicals. *J. Phys. Chem. A* **2021**, *125*, 5103−5116.

(52) Passos, M. O.; Lins, I. A.; Venâncio, M. F.; Alves, T. V. Differences in the torsional anharmonicity between reactant and transition state: The case of 3-butenal + H abstraction reactions. *Phys. Chem. Chem. Phys.* **2021**, *23*, 25414−25423.

(53) Alecu, I. M.; Gao, Y.; Marshall, P. Experimental and computational studies of the kinetics of the reaction of hydrogen peroxide with the amidogen radical. *J. Chem. Phys.* **2022**, *157*, 014304.

(54) Duan, Y.; Ren, Z.; Huang, Z.; Han, D. Theoretical study on isomerization, decomposition and ring-closure reaction kinetics of methyl pentanoate radicals. *Combust. Flame* **2022**, *237*, 111848.

(55) Ferro-Costas, D.; Cordeiro, M. N. D. S.; Truhlar, D. G.; Fernández-Ramos, A. Q2DTor: A program to treat torsional anharmonicity through coupled pair torsions in flexible molecules. *Comput. Phys. Commun.* **2018**, *232*, 190−205.

(56) Ferro-Costas, D.; Mosquera-Lois, I.; Fernández-Ramos, A. TorsiFlex: An automatic generator of torsional conformers. Application to the twenty proteinogenic amino acids. *J. Cheminf.* **2021**, *13*, 100.

(57) Ferro-Costas, D.; Fernández-Ramos, A. A combined systematic-stochastic algorithm for the conformational search in flexible acyclic molecules. *Front. Chem.* **2020**, *8*, 16.

(58) Zheng, J.; Mielke, S. L.; Bao, J. L.; Meana-Pañeda, R.; Clarkson, K. L.; Truhlar, D. G.*MSTor Computer Program*, Version 2017; University of Minnesota: Minneapolis, MN, 2017.

(59) Baker, J.; Kessi, A.; Delley, B. The generation and use of delocalized internal coordinates in geometry optimization. *J. Chem. Phys.* **1996**, *105*, 192−212.

(60) Vereecken, L.; Glowacki, D. R.; Pilling, M. J. Theoretical chemical kinetics in tropospheric chemistry: Methodologies and applications. *Chem. Rev.* **2015**, *115*, 4063−4114.

(61) Dzib, E.; Merino, G. The hindered rotor theory: A review. *Wiley Interdiscip. Rev.: Comput. Mol. Sci.* **2022**, *12*, No. e1583.

(62) Sharma, S.; Raman, S.; Green, W. H. Intramolecular hydrogen migration in alkylperoxy and hydroperoxyalkylperoxy radicals: Accurate treatment of hindered rotors. *J. Phys. Chem. A* **2010**, *114*, 5689−5701.

(63) Ayala, P. Y.; Schlegel, H. B. Identification and treatment of internal rotation in normal mode vibrational analysis. *J. Chem. Phys.* **1998**, *108*, 2314−2325.

(64) Jamróz, M. H. Vibrational energy distribution analysis (VEDA): Scopes and limitations. *Spectrochim. Acta, Part A* **2013**, *114*, 220−230.

(65) Billeter, S. R.; Turner, A. J.; Thiel, W. Linear scaling geometry optimisation and transition state search in hybrid delocalised internal coordinates. *Phys. Chem. Chem. Phys.* **2000**, *2*, 2177−2186.

(66) Wilson, E. B., Jr.; Decius, J. C.; Cross, P. C.*Molecular Vibrations: The Theory of Infrared and Raman Vibrational Spectra*; McGraw-Hill: London, 1955.

(67) Ochterski, J. W.Vibrational analysis in Gaussian. https://gaussian.com/vib/ (accessed Sept 19, 2022).

(68) Bao, J. L.; Truhlar, D. G. Variational transition state theory: Theoretical framework and recent developments. *Chem. Soc. Rev.* **2017**, *46*, 7548−7596.

(69) Peng, C.; Ayala, P. Y.; Schlegel, H. B.; Frisch, M. J. Using redundant internal coordinates to optimize equilibrium geometries and transition states. *J. Comput. Chem.* **1996**, *17*, 49−56.

(70) Ferro-Costas, D.; Truhlar, D. G.; Fernández-Ramos, A. Pilgrim: A thermal rate constant calculator and a chemical kinetics simulator. *Comput. Phys. Commun.* **2020**, *256*, 107457.

(71) Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Petersson, G. A.; Nakatsuji, H.; Li, X.; Caricato, M.; Marenich, A. V.; Bloino, J.; Janesko, B. G.; Gomperts, R.; Mennucci, B.; Hratchian, H. P.; Ortiz, J. V.; Izmaylov, A. F.; Sonnenberg, J. L.; Williams-Young, D.; Ding, F.; Lipparini, F.; Egidi, F.; Goings, J.; Peng, B.; Petrone, A.; Henderson, T.; Ranasinghe, D.; Zakrzewski, V. G.; Gao, J.; Rega, N.; Zheng, G.; Liang, W.; Hada, M.; Ehara, M.; Toyota, K.; Fukuda, R.; Hasegawa, J.; Ishida, M.; Nakajima, T.; Honda, Y.; Kitao, O.; Nakai, H.; Vreven, T.; Throssell, K.; Montgomery, J. A., Jr.; Peralta, J. E.; Ogliaro, F.; Bearpark, M. J.; Heyd, J. J.; Brothers, E. N.; Kudin, K. N.; Staroverov, V. N.; Keith, T. A.; Kobayashi, R.; Normand, J.; Raghavachari, K.; Rendell, A. P.; Burant, J. C.; Iyengar, S. S.; Tomasi, J.; Cossi, M.; Millam, J. M.; Klene, M.; Adamo, C.; Cammi, R.; Ochterski, J. W.; Martin, R. L.; Morokuma, K.; Farkas, O.; Foresman, J. B.; Fox, D. J.*Gaussian 16*, Revision B.01; Gaussian, Inc.: Wallingford CT, 2016.

(72) Zhao, Y.; Truhlar, D. G. The M06 suite of density functionals for main group thermochemistry, thermochemical kinetics, noncovalent interactions, excited states, and transition elements: Two new functionals and systematic testing of four M06-class functionals and 12 other functionals. *Theor. Chem. Acc.* **2008**, *120*, 215−241.

(73) Weigend, F.; Ahlrichs, R. Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy. *Phys. Chem. Chem. Phys.* **2005**, *7*, 3297−3305.

(74) Chai, J.-D.; Head-Gordon, M. Long-range corrected hybrid density functionals with damped atom−atom dispersion corrections. *Phys. Chem. Chem. Phys.* **2008**, *10*, 6615−6620.

(75) Kendall, R. A.; Dunning, T. H.; Harrison, R. J. Electron affinities of the first-row atoms revisited. Systematic basis sets and wave functions. *J. Chem. Phys.* **1992**, *96*, 6796−6806.
(76) Glowacki, D. R.; Liang, C.-H.; Morley, C.; Pilling, M. J.; Robertson, S. H. MESMER: An open-source master equation solver for multi-energy well reactions. *J. Phys. Chem. A* **2012**, *116*, 9545−9560.
(77) Gao, Y.; He, T.; Li, X.; You, X. Effect of hindered internal rotation treatments on predicting the thermodynamic properties of alkanes. *Phys. Chem. Chem. Phys.* **2019**, *21*, 1928−1936.

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